WP15 Solutions

102. $F(x,y,z) = y\hat{i} + (x+z)\hat{j} - y\hat{k}$ is conservative?

No. consider the paths
$r_1(t) = \langle t,t,t\rangle$ and
$r_2(t) = \langle t, t^2, t\rangle$

$$\int_{C_1} F\cdot dr = \int_0^1 \langle t, 2t, -t\rangle \cdot \langle 1,1,1\rangle dt$$
$$= \int_0^1 2t\, dt$$
$$= t^2 \Big|_0^1$$
$$= 1$$

$$\int_{C_2} F dr = \int_0^1 \langle t^2, 2t, -t^2\rangle \langle 1, 2t, 1\rangle dt$$
$$= \int_0^1 4t^2\, dt$$
$$= \tfrac{4}{3}t^3 \Big|_0^1$$
$$= 4/3$$

so $\int_{C_1} F\cdot dr \neq \int_{C_2} F\cdot dr$

113. $\oint_C (2y\,dx + 2x\,dy)$ C: line from $(0,0)$ to $(4,4)$

$f(x,y) = 2xy$
$r(t) = \langle t,t\rangle$ for $t \in [0,4]$
so $\int_C \nabla f \cdot dr$
$$= f(r(4)) - f(r(0))$$
$$= f(4,4) - f(0,0)$$
$$= 32 - 0$$
$$= 32$$

116. $F(x,y) = (12xy)\hat{i} + 6(x^2+y^2)\hat{j}$

test $r_1(t) = \langle t,t\rangle$ and $r_2(t) = \langle t, t^2\rangle$ on $(0,1)$

$$\int_{C_1} F\cdot dr = \int_0^1 \langle 12t^2, 12t^2\rangle \langle 1,1\rangle dt$$
$$= \int_0^1 24t^2\, dt$$
$$= 8t^3 \Big|_0^1$$
$$= 8$$

$$\int_{C_2} F\cdot dr = \int_0^1 \langle 12t^3, 6(t^2+t^4)\rangle \langle 1, 2t\rangle dt$$
$$= \int_0^1 24t^3 + 12t^5\, dt$$
$$= 6t^4 + 2t^6 \Big|_0^1$$
$$= 8$$

Conservative field

$$\int 12xy\, dx = 6x^2y + g(y)$$
$$6x^2 + g_y = 6(x^2+y^2)$$
$$g_y = 6y^2$$
$$g(y) = 2y^3$$

so $f(x,y) = 6x^2y + 2y^3$ is the potential function

119. $F(x,y,z) = (ye^z)\hat{i} + (xe^z)\hat{j} + (xye^z)\hat{k}$

let $r_1(t) = \langle t,t,t \rangle$ for $t \in [0,1]$ and $r_2(t) = \langle t,t,t^2 \rangle$ for $t \in [0,1]$

$$\int_{C_1} F \cdot dr = \int_0^1 \langle te^t, te^t, t^2e^t \rangle \cdot \langle 1,1,1 \rangle \, dt$$
$$= \int_0^1 (2t+t^2)\, e^t \, dt$$
$$= (2t+t^2)e^t - (2+2t)e^t + 2e^t \Big|_0^1$$
$$= (3e - 4e + 2e) - (0 - 2 + 2)$$
$$= e$$

$$\int_{C_2} F \cdot dr = \int_0^1 \langle te^{t^2}, te^{t^2}, t^2e^{t^2} \rangle \cdot \langle 1,1,2t \rangle \, dt$$
$$= \int_0^1 (2t + 2t^3) e^{t^2} \, dt$$
$$= (1+t^2)e^{t^2} - e^{t^2} \Big|_0^1$$
$$= (2e - e) - (1-1)$$
$$= e$$

Since these are the same this is a conservative vector field

to get potential

$$\int ye^z \, dx = xye^z + g(y,z)$$

so $xe^z + g_y = xe^z$ so $g_y = 0$

$f_z = xye^z + h'(z) \Rightarrow h'(z) = 0$

so $f(x,y,z) = xye^z$ is the potential function

126. $f(x,y,z) = \cos(\pi x) + \sin(\pi y) - xyz$ C: path from $(1, \frac{1}{2}, 2)$ to $(2,1,-1)$

$r(t_i) = (1, \frac{1}{2}, 2)$ and $r(t_f) = (2,1,-1)$

so $\int_C \nabla f \cdot dr = f(r(t_f)) - f(r(t_0))$

$$= f(2,1,-1) - f(1, \tfrac{1}{2}, 2)$$
$$= [\cos(2\pi) + \sin \pi - 2(1)(-1)] - [\cos \pi + \sin \tfrac{\pi}{2} - 1(\tfrac{1}{2})(2)]$$
$$= (1 + 0 + 2) - (-1 + 1 - 1)$$
$$= 4$$

143. $F = -y\hat{i} + x\hat{j}$ around and across $r_1(t) = (a\cos t)\hat{i} + (a\sin t)\hat{j}$;
$0 \le t \le \pi$ and $r_2(t) = t\hat{i}$ $-a \le t \le a$

$$\text{Flux} = \int_C F \cdot N \, dS$$
$$= \int_0^{\pi} \langle -a\sin t, a\cos t\rangle \cdot \langle a\cos t, a\sin t\rangle dt + \int_{-a}^{a} \langle 0, t\rangle \cdot \langle 1, 0\rangle \, dt$$
$$= \int_0^{\pi} (-a^2 \sin t\cos t + a^2 \sin t\cos t) dt + \int_{-a}^{a} (0+0) \, dt$$
$$= \int_0^{\pi} 0 \, dt + \int_{-a}^{a} 0 \, dt$$
$$\text{Flux} = 0$$

$$\text{Circulation} = \int_C F \cdot T \, dS$$
$$= \int_0^{\pi} \langle -a\sin t, a\cos t\rangle \cdot \langle -a\sin t, a\cos t\rangle dt + \int_{-a}^{a} \langle 0, t\rangle \cdot \langle 1, 0\rangle dt$$
$$= \int_0^{\pi} (a^2 \sin^2 t + a^2 \cos^2 t) dt + \int_0^{a} (0+0) \, dt$$
$$= \int_0^{\pi} a^2 \, dt + 0$$
$$= a^2 t \Big|_0^{\pi}$$

$$\text{Circulation} = \pi a^2$$